

Market-based Value-at-Risk Constraints in Hedge Funds

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Abstract—This study explores the impact of a stochastic Value-at-Risk (VaR) constraint, tied to a market-based benchmark, on the portfolio management strategies of a hedge fund manager characterized by an S-shaped utility function under an incentive contract that includes management and performance fees. We integrate the market-based VaR constraint into the hedge fund manager's utility maximization framework, which accounts for dynamic market conditions and the manager's relative performance. Employing stochastic control theory, we derive the optimal portfolio strategy that balances risk and reward while adhering to the VaR limit. Our findings reveal how the market-based VaR constraint influences the manager's risk-taking behaviors, adjusting asset exposure in response to market fluctuations and performance metrics. This research contributes to theoretical portfolio management literature and offers practical insights for designing effective market-based risk management strategies.

Index Terms—Martingale Method, stochastic Value-at-Risk constraint

I. INTRODUCTION

In the domain of continuous-time portfolio selection, the classical framework primarily focuses on the maximization of expected utility, where portfolio choices are steered towards optimizing terminal wealth under specific constraints. Traditionally, this optimization is formulated as:

$$\begin{aligned} & \max_{X_T \in \mathcal{V}} \mathbb{E}[U(X_T)] \\ & \text{s.t. } \mathbb{E}[\xi X_T] \leq x_0 \text{ a.s.} \end{aligned} \quad (1)$$

Here, U denotes a Hyperbolic Absolute Risk Aversion (HARA) utility function, X_T represents the terminal wealth, and ξ is a pricing kernel with $x_0 \in \mathbb{R}$ serving as the budget constraint's upper limit. This classical model, foundational to seminal work of (1), has evolved through the integration of various utility forms that cater to different financial behaviors and market conditions.

Recent advancements in utility theory, such as those presented by (2), propose a piece-wise HARA utility, which generalizes traditional utility functions to accommodate non-standard preferences within portfolio optimization frameworks. This approach allows for the customization of utility functions to reflect more complex investor preferences and scenarios.

Furthermore, a significant shift in the literature has been towards integrating behavioral economics concepts into these models. For instance, the S-shaped utility, rooted in behavioral economics, captures the nuances of human risk behavior more accurately than traditional utilities. In line with this thought, we introduce a novel concept of a market-based stochastic Value-at-Risk (VaR) constraint. This approach is inspired by the work of (3), who employed an S-shaped utility alongside a singular VaR constraint, demonstrating how these models can yield optimal portfolios that are attuned to behavioral risk assessments.

This paper aims to extend these models by exploring the implications of market-based stochastic VaR constraints in portfolio optimization, offering a comprehensive analysis of portfolios under various scenarios and utility frameworks. Our findings provide a systematic derivation of closed-form solutions for optimal portfolio policies under these complex conditions.

The structure of this paper is organized as follows: Section II outlines the establishment of the optimal control problem in a continuous-time setting and S-shape utility. Section ?? is dedicated to deriving the closed-form solutions for a single investor's optimization problem with market-based stochastic VaR constraint. Section III gives a simple example of hedge fund managers' optimal strategy with an incentive contract. Section IV is the main part of our paper, involving the market-based stochastic VaR constraints applied by hedge fund managers with a contract. The subsequent sections will delve deeper into the application of these strategies in hedge fund, followed by a conclusive discussion and detailed proofs provided in the appendix.

II. MODEL SETTING

A. Market model

In our analysis, we employ the canonical Black-Scholes framework to model the price dynamics of $m \geq 1$ risky assets, adhering to the assumptions and formulations that underpin classical derivative pricing theories. The financial market considered here consists of m risky assets, each denoted by i , where $i = 1, \dots, m$. The dynamics of the price process $S_{i,t}$ for each asset are driven by an l -dimensional standard Brownian motion $\mathbf{W}t = (W1, t, \dots, Wl, t)'$, which reflects various systematic risks. Specifically, the price of

the i -th asset evolves according to the stochastic differential equation:

$$dS_{i,t} = S_{i,t} \left(\mu_{i,t} dt + \sum_{j=1}^l \sigma_{i,j,t} dW_{j,t} \right), ; 0 \leq t \leq T. \quad (2)$$

where $\mu_{i,t}$ represents the instantaneous expected return of the i -th asset at time t , and $\sigma_{i,j,t}$ quantifies the volatility or sensitivity of the i -th asset's price to the j -th Brownian motion.

Additionally, the financial market includes a risk-free asset, whose price process $S_{0,t}$ evolves according to:

$$dS_{0,t} = r_t S_{0,t} dt, ; 0 \leq t \leq T, \quad (3)$$

where r_t is the risk-free rate, providing a baseline for assessing the performance of risky investments.

The market model we consider is a complete market, where the number of risky assets m equals the number of sources of randomness l , thus enabling perfect hedging of all market risks. In such a market, the dynamics of each asset i , where $i = 1, \dots, m$, are described by:

$$dS_{i,t} = S_{i,t} \left(\mu_i dt + \sum_{j=1}^m \sigma_{i,j} dW_{j,t} \right), 0 \leq t \leq T,$$

where μ_i represents the expected return rate and $\sigma_{i,j}$ denotes the constant volatility of the i -th asset with respect to the j -th Brownian motion. This setup allows the derivation of closed-form solutions and provides significant insights into financial market behaviors, leveraging the theoretical framework established by (4) and (5). The assumption of constant market parameters simplifies the analysis while capturing essential features relevant to financial markets.

Hence, the volatility matrix σ plays a crucial role, representing the sensitivities of m risky assets to various risk factors. It is defined as $\sigma = (\sigma_{i,j})_{1 \leq i, j \leq m}$, a square matrix that underlies the diffusion terms of the asset prices' dynamics.

A fundamental regularity condition imposed on σ ensures mathematical tractability and robustness of the model, particularly important for the pricing of derivatives. This condition is expressed as:

$$\eta^\top (\sigma \sigma^\top) \eta \geq \varepsilon \|\eta\|_2^2, \quad (4)$$

where $\|\eta\|_2$ is the Euclidean norm of η , and $\varepsilon > 0$ is a constant ensuring that the matrix $\sigma \sigma^\top$ is uniformly positive definite. This guarantees the invertibility of the matrix, which is essential for the formulation of the pricing kernel and ensuring no-arbitrage conditions within the market model.

The invertibility of σ and its transpose, along with the positive definiteness of $\sigma \sigma^\top$ (denoted as Σ), also imply market completeness. In a complete market, every contingent claim can be perfectly hedged. This concept is extensively discussed in foundational texts like (6), which provide a comprehensive mathematical framework for such financial models. This setup aligns with established financial models including the Heath-Jarrow-Morton framework for interest rates.

To encapsulate the portfolio dynamics under consideration, let $\pi_t = (\pi_{1,t}, \dots, \pi_{m,t})^\top$ represent the vector of monetary amounts invested in each risky asset at time t . The total value of the portfolio, denoted by $X_{t0 \leq t \leq T}$, is given by the solution to the stochastic differential equation:

$$dX_t = (rX_t + \pi_t^\top (\mu - r\mathbf{1}_m)) dt + \pi_t^\top \sigma d\mathbf{W}_t, X_0 = x_0. \quad (5)$$

In this equation, μ is the vector of expected returns on the assets, and $\mathbf{1}_m$ is a vector of ones multiplied by the risk-free rate r , adjusting the returns for the risk-free borrowing cost. The term $\pi_t^\top \sigma d\mathbf{W}_t$ reflects the uncertainty in returns due to exposure to systematic risk factors.

An admissible portfolio strategy $\pi = \{\pi_t\}_{0 \leq t \leq T}$ must be progressively measurable with respect to the filtration generated by $\mathbf{W}_t$ and satisfy $\int_0^T \|\pi_t\|_2^2 dt < \infty$ almost surely, ensuring the financial viability of the strategy throughout the trading horizon.

Within our financial model framework, the pricing kernel, denoted as $\{\xi_t\}_{0 \leq t \leq T}$, is instrumental in the valuation of risky assets and derives its dynamics from the preferences of economic agents and the uncertainties inherent in the market. The pricing kernel is defined to ensure no-arbitrage conditions and equilibrium in the market, connecting the risk-neutral probabilities with the actual probabilities of financial outcomes.

Formally, the pricing kernel can be represented through its relationship with the risk-free rate r and the market's risk factors. The dynamics of the pricing kernel are given by:

$$d\xi_t := -\xi_t (r dt + \theta^\top d\mathbf{W}_t), \quad (6)$$

or equivalently:

$$\xi_t := \exp \left\{ -\left(r + \frac{1}{2} \|\theta\|_2^2\right)t - \theta^\top \mathbf{W}_t \right\}, 0 \leq t \leq T. \quad (7)$$

where $\theta := \sigma^{-1}(\mu - r\mathbf{1}_m)$ represents the market price of risk at time t . This vector measures the additional returns per unit of risk that investors demand over the risk-free rate, reflecting the risk preferences of market participants. The term $\theta^\top d\mathbf{W}_t$ captures the adjustments to the discount factor due to changes in the underlying risk factors.

B. S-shaped utility

In refining the discussion on the behavioral strategies of hedge fund managers under risk, it's essential to consider their goal of maximizing expected utility within a specified time frame $[0, T]$, while contending with loss aversion. Traditional studies in this area typically focus on strictly concave utility functions which assume a consistent attitude towards risk across different financial outcomes. However, the introduction of prospect theory by (7) offers a pivotal deviation from this norm. Their theory posits that individuals exhibit both loss aversion and risk aversion, significantly influenced by a relative reference point, encapsulating these behaviors within an S-shaped utility function, which starkly contrasts with the classical concave utility models.

In our model, we assume the manager has an S-shape utility:

$$\hat{U}(x) = \begin{cases} k_1(x - B_1)^{\gamma_1}, & x \geq B_1; \\ -k_2(B_1 - x)^{\gamma_2}, & x < B_1. \end{cases} \quad (8)$$

where B_1 is the reference point, $k_1 > 0$ and $k_2 > 0$ and $1 < \gamma_1 \leq 1$ and $1 < \gamma_2 < 1$ respectively measure the degree of risk aversion and risk seeking. The manager compares her payoff and loss with the reference level B_1 . She becomes loss-averse (risk-averse) under (over) the reference level.

III. OPTIMIZATION PROBLEM IN HEDGE FUNDS

A. Incentive contract in the hedge fund in the US market

As an application, we give a classic structure of hedge fund management; see (8). The manager's preference $\hat{U}$ is an S-shaped utility, and the payoff function Θ is

$$\Theta(X_T) = \omega X_T + \alpha(1 - \omega)X_T + \beta(1 - \omega) \max\{X_T - B_2, 0\}, \\ X_T \geq bx_0 e^{rT}, \quad (9)$$

where $\omega \in (0, 1)$ is the proportion of the manager in the fund wealth, $\alpha > 1$ is the management fee, $\beta > \psi$ is the incentive rate and $bx_0 > 0$ is the liquidation boundary. For convenience, we denote $\underline{x} := \Theta(bx_0 e^{rT})$. B_2 is the benchmark wealth. It can be set as a c -multiple of the riskless investment benchmark, i.e., $B_2 = cx_0 e^{rT}$, $c > 0$. Further, the parameter in the PHARA utility can be computed according to the financial market. In the US, for an investment period $T = 10$, a typical setting is $\omega = 0.1$, $\alpha = 0.2$, $\beta = 0.4$, $b = 0.7$, $c = 2$. The calibration of the S-shaped utility can be referred to (9).

B. Problem 1: an S-shaped utility and a contract

The investor solves an optimization problem to conduct portfolio selection:

$$\max_{\pi \in \mathcal{V}} \mathbb{E}[U(X_T)] = \max_{\pi \in \mathcal{V}} \mathbb{E}[\hat{U} \circ \Theta(X_T)], \quad (10)$$

where $U := \hat{U} \circ \Theta$ is the objective function (the composed utility of the manager's preference and payoff) and $\mathcal{V}$ is the collection of all the admissible portfolios.

The concave envelope plays an important role in portfolio selection problems with non-concave utilities. Indeed, if ξ_T in (7) has a continuous distribution, Problem (10) has the same optimal solution as the following problem with the utility U replaced by the concave envelope U^{**} :

$$\max_{\pi \in \mathcal{V}} \mathbb{E}[U^{**}(X_T)]. \quad (11)$$

Thus, we can study U^{**} instead of U and later obtain the same optimal portfolio for both Problems (10) and (11).

C. Problem modelling

Here we use the S-shape utility function $\hat{U}$ in (8) and incentive contract in (9). There are two cases of the objective function, which depends on the choice of reference level B_1 of $\hat{U}$ (8), benchmark B_2 and the payoff function $\Theta(\cdot)$. Note that the payoff function (9) is strictly increasing, hence its inverse function $\Theta^{-1}(\cdot)$ is well-defined on $[\Theta(bx_0 e^{rT}), \infty)$. We need to compare the two points: $x = \Theta^{-1}(B_1)$ and $x = B_2$.

$\Theta^{-1}(B_1)$ is the *effective reference level* for the manager based on the S-shape utility.

Analytically, we have

(1) if $B_1 \geq \Theta(B_2)$:

$$U(x) = \begin{cases} k_1[\omega x + \alpha(1 - \omega)x + \beta(1 - \omega)(x - B_2) - B_1]^{\gamma_1}, & x > \Theta^{-1}(B_1), \\ -k_2[B_1 - (\omega x + \alpha(1 - \omega)x + \beta(1 - \omega)(x - B_2))]^{\gamma_2}, & B_2 \leq x \leq \Theta^{-1}(B_1), \\ -k_2[B_1 - (\omega x + \alpha(1 - \omega)x)]^{\gamma_2}, & x \leq B_2. \end{cases} \quad (12)$$

(2) if $B_1 < \Theta(B_2)$:

$$U(x) = \begin{cases} k_1[\omega x + \alpha(1 - \omega)x + \beta(1 - \omega)(x - B_2) - B_1]^{\gamma_1}, & x > B_2, \\ k_1[\omega x + \alpha(1 - \omega)x - B_1]^{\gamma_1}, & \Theta^{-1}(B_1) \leq x \leq B_2, \\ -k_2[B_1 - (\omega x + \alpha(1 - \omega)x)]^{\gamma_2}, & x \leq \Theta^{-1}(B_1). \end{cases} \quad (13)$$

Although the utility function, U is non-concave, it is a type of PHARA utility, so we can study its concave envelope U^{**} . Lemma 1 discusses the existence of two tangent lines and establishes the concave envelope of U , and is used to obtain the optimal terminal wealth X_T^* in Theorem 1.

Lemma 1. (1) If $B_1 < \Theta(B_2)$, there exists a tangent point $(1 + c_1)\frac{B_1}{\alpha_1} \in (\frac{B_1}{\alpha_1}, B_2]$ deduced from $(\underline{x}, U(\underline{x}))$ to $U|_{(\frac{B_1}{\alpha_1}, B_2]}$, which satisfies

$$\frac{U((1 + c_1)\frac{B_1}{\alpha_1}) - U(\underline{x})}{(1 + c_1)\frac{B_1}{\alpha_1} - \underline{x}} = U'((1 + c_1)\frac{B_1}{\alpha_1}). \quad (14)$$

Define κ_1 as the slope of the tangent line:

$$\kappa_1 \triangleq U'((1 + c_1)\frac{B_1}{\alpha_1}). \quad (15)$$

(1)' If $B_1 \geq \Theta(B_2)$, there exists a tangent point $(1 + c_1)\frac{B_1}{\alpha_1} \in (\frac{B_1}{\alpha_1}, \infty)$ deduced from $(\underline{x}, U(\underline{x}))$ to $U|_{(\frac{B_1}{\alpha_1}, \infty)}$, which satisfies

$$\frac{U((1 + c_1)\frac{B_1}{\alpha_1}) - U(\underline{x})}{(1 + c_1)\frac{B_1}{\alpha_1} - \underline{x}} = U'((1 + c_1)\frac{B_1}{\alpha_1}). \quad (16)$$

Define κ_1 as the slope of the tangent line:

$$\kappa_1 \triangleq U'((1 + c_1)\frac{B_1}{\alpha_1}). \quad (17)$$

(2) If $B_1 < \Theta(B_2)$, there exists tangent points $(1 + c_2)\frac{B_1}{\alpha_1} \in (\frac{B_1}{\alpha_1}, B_2]$ and $(1 + c_3)\frac{\beta(1 - \omega)B_2 + B_1}{\alpha_2} \in (B_2, +\infty)$, which satisfy

$(\underline{x}, U(\underline{x}))$ to $U|_{(\frac{B_1}{\alpha_1}, B_2)}$, which satisfies

$$\begin{aligned} & \frac{U((1+c_2)\frac{B_1}{\alpha_1}) - U((1+c_3)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2})}{(1+c_2)\frac{B_1}{\alpha_1} - (1+c_3)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2}} = U'((1+c_2)\frac{B_1}{\alpha_1}) \\ & = U'((1+c_3)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2}). \end{aligned} \quad (18)$$

Define κ_2 as the slope of the tangent line:

$$\kappa_2 \triangleq U'((1+c_2)\frac{B_1}{\alpha_1}). \quad (19)$$

(3) If $B_1 < \Theta(B_2)$ and $\kappa_1 \leq \kappa_2$, there exists a tangent point $(1+c_4)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} \in (B_2, +\infty)$ deduced from $(\underline{x}, U(\underline{x}))$ to $U|_{(B_2, +\infty)}$, which satisfies

$$\begin{aligned} & \frac{U((1+c_4)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2}) - U(\underline{x})}{(1+c_4)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} - \underline{x}} \\ & = U'((1+c_4)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2}). \end{aligned} \quad (20)$$

Define κ_3 as the slope of the tangent line:

$$\kappa_3 \triangleq U'((1+c_4)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2}). \quad (21)$$

where $\alpha_1 := \omega + \alpha(1-\omega)$, $\alpha_2 := \omega + (\alpha + \beta)(1-\omega)$

As a result, we obtain the concave envelope in all cases:

(1) if $B_1 \geq \Theta(B_2)$:

$$U^{**}(x) = \begin{cases} \kappa_1(x - \underline{x}) + U(\underline{x}), & \underline{x} \leq x < (1+c_1)\frac{B_1}{\alpha_1}, \\ U(x), & (1+c_1)\frac{B_1}{\alpha_1} \leq x, \end{cases} \quad (22)$$

There exists a unique tangent point $(1+c_1)\Theta^{-1}(B_1)$ the slope κ_1 is defined by

(2) if $B_1 < \Theta(B_2)$, and $\kappa_1 > \kappa_2$:

$$U^{**}(x) = \begin{cases} \kappa_1(x - \underline{x}) + U(\underline{x}), & \underline{x} \leq x < (1+c_1)\frac{B_1}{\alpha_1} \\ U(x), & (1+c_1)\frac{B_1}{\alpha_1} \leq x < (1+c_2) \\ \kappa_2(x - ((1+c_2)\frac{B_1}{\alpha_1})) + U((1+c_2)\frac{B_1}{\alpha_1}), & (1+c_2)\frac{B_1}{\alpha_1} \leq x < (1+c_3)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} \\ U(x), & (1+c_3)\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} \leq x \end{cases} \quad (23)$$

(3) if $B_1 < \Theta(B_2)$, and $\kappa_1 \leq \kappa_2$:

$$U^{**}(x) = \begin{cases} \kappa_3(x - \underline{x}) + U(\underline{x}), & \underline{x} \leq x < \frac{B_1}{\alpha_1}, \\ U(x), & (1+c_1)\frac{B_1}{\alpha_1} \leq x, \end{cases} \quad (24)$$

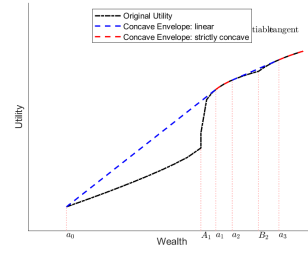


Fig. 1: Utility with its concave envelope under the case: $B_1 < \Theta(B_2)$ and $\kappa_1 > \kappa_2$.

Example 1. We provide an example under the preceding case (2). Figure 1 illustrates the utility function with its concave envelope. In this case, we have partition points: $a_0 = \underline{x}$, $a_1 = (1+c_1)A_1 = 19.5509$, $a_2 = (1+c_2)A_2 = 21.2520$, $a_3 = (1+c_3)A_3 = 26.1221$, reference points: $A_0 = a_0 = 4$, $A_1 = A_2 = \frac{B_1}{\alpha_1} = 18$, $A_3 = \frac{\beta(1-\omega)B_2+B_1}{\alpha_2} = 21.8961$, other parameters: $\omega = 0.2$, $\alpha = 2\%$, $\gamma_1 = 0.2$, $\gamma_2 = 0.8$, $\beta = 0.5$, $\underline{x} = 4$, $B_1 = 3.888$, $B_2 = 24$. The concave envelope is the smallest concave function dominating the utility function. If the concave envelope is strictly concave on some interval, the concave envelope overlaps the original utility function on the interval.

We can solve the optimal terminal wealth of the Problem 11 from the concave envelope. The results are presented in Theorem 1

D. Problem-solving: Optimal portfolio

We obtain the optimal portfolio vector from the unified formula in (2).

Theorem 1. The closed-form optimal wealth process X^* and the optimal portfolio π^* for Problems (10) and (11) are given by:

(1) If $B_1 \geq \Theta(B_2)$, then the optimal terminal wealth X_T^* is

$$X_T^* = \underline{x} \mathbb{1}_{\{\nu^* \xi_T \geq \kappa_1\}} + (c_1 (\frac{\nu^* \xi_T}{\kappa_1})^{\frac{1}{\gamma_1-1}} + 1) \frac{B_1 + \beta(1-\omega)B_2}{\alpha_2} \mathbb{1}_{\{\nu^* \xi_T < \kappa_1\}}, \quad (25)$$

and the optimal wealth amount at time t is

$$\begin{aligned} X_t^* &= \xi_t^{-1} \mathbb{E}[\xi_T X_T^* | \mathcal{F}_t] \\ &= e^{-r(T-t)} \underbrace{\underline{x} \Phi(d_1(\frac{\kappa_1}{\nu^* \xi_t}))}_{\text{first-order risk aversion}} \\ &\quad + e^{-r(T-t)} \underbrace{\left\{ \frac{\beta(1-\omega)B_2+B_1}{\alpha_2} [1 - \Phi(d_1(\frac{\kappa_1}{\nu^* \xi_t}))] \right\}}_{\text{loss aversion}} \\ &\quad + e^{-r(T-t)} \underbrace{\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} c_1 \times \frac{\Phi'(d_1(\frac{\kappa_1}{\nu^* \xi_t}))}{\Phi'(d_2(\frac{\kappa_1}{\nu^* \xi_t}))}}_{\text{risk seeking}} \\ &\quad \times [1 - \Phi(d_2(\frac{\kappa_1}{\nu^* \xi_t}))] \\ &\triangleq X_t^D + X_t^A + X_t^R. \end{aligned} \quad (26)$$

where ν^* is a unique positive number that satisfies $\mathbb{E}[\xi_T X_T^*] = x_0$, and $\Phi(\cdot)$ is the standard normal cumulative distribution function.

The optimal amount allocated in the risky asset, i.e., the optimal portfolio π^* is

$$\begin{aligned} \pi_t^* &= \underbrace{\frac{(\sigma^T)^{-1}\theta}{1-\gamma_1} X_t^*}_{\text{Merton term}} \\ &+ \underbrace{\frac{e^{-r(T-t)}}{\sqrt{T-t}} \frac{(\sigma^T)^{-1}\theta}{\|\theta\|_2} [(1+c_1) \frac{\beta(1-\omega)B_2+B_1}{\alpha_2} - \underline{x}]}_{\text{risk seeking}} \\ &\quad \underbrace{\Phi'(d_1(\frac{\kappa_1}{\nu^*\xi_t}))}_{\text{risk seeking}} \\ &- \underbrace{e^{-r(T-t)} \frac{\beta(1-\omega)B_2+B_1}{\alpha_2} [1 - \Phi(d_1(\frac{\kappa_1}{\nu^*\xi_t}))]}_{\text{loss aversion}} \\ &- \underbrace{e^{-r(T-t)} \frac{(\sigma^T)^{-1}\theta}{1-\gamma_1} \{ \underline{x} \Phi(d_1(\frac{\kappa_1}{\nu^*\xi_t})) \}}_{\text{first-order risk aversion}} \\ &\triangleq \pi_t^{(1)} + \pi_t^{(2)} + \pi_t^{(3)} + \pi_t^{(4)}. \end{aligned} \quad (27)$$

where $\Phi'(\cdot)$ is the standard normal probability distribution function.

(2) If $B_1 < \Theta(B_2)$, and $\kappa_1 > \kappa_2$, then the optimal terminal wealth X_T^* is

$$\begin{aligned} X_T^* &= \underline{x} \mathbb{1}_{\{\nu^*\xi_T \geq \kappa_1\}} + [c_1 (\frac{\nu^*\xi_T}{\kappa_1})^{\frac{1}{\gamma_1-1}} + 1] \frac{B_1}{\alpha_1} \mathbb{1}_{\{\kappa_2 < \nu^*\xi_T < \kappa_1\}} \\ &+ [c_3 (\frac{\nu^*\xi_T}{\kappa_2})^{\frac{1}{\gamma_1-1}} + 1] \frac{\beta(1-\omega)B_2+B_1}{\alpha_2} \mathbb{1}_{\{\nu^*\xi_T \leq \kappa_2\}}, \end{aligned} \quad (28)$$

and the optimal wealth process X^* is

$$\begin{aligned} X_t^* &= \xi_t^{-1} \mathbb{E}[\xi_T X_T^* | \mathcal{F}_t] \\ &= e^{-r(T-t)} \underbrace{\underline{x} \Phi(d_1(\frac{\kappa_1}{\nu^*\xi_t}))}_{\text{first-order risk aversion}} \\ &+ e^{-r(T-t)} \underbrace{\{ \frac{B_1}{\alpha_1} [\Phi(d_1(\frac{\kappa_2}{\nu^*\xi_t})) - \Phi(d_1(\frac{\kappa_1}{\nu^*\xi_t}))] + \beta(1-\omega)B_2 + B_1 \frac{1}{\alpha_2 [1-\Phi(d_1(\frac{\kappa_2}{\nu^*\xi_t}))]} \}}_{\text{loss aversion}} \\ &+ e^{-r(T-t)} \underbrace{\{ \frac{B_1}{\alpha_1} c_1 \times \frac{\Phi'(d_1(\frac{\kappa_1}{\nu^*\xi_t}))}{\Phi'(d_2(\frac{\kappa_1}{\nu^*\xi_t}))} \times [\Phi(d_2(\frac{\kappa_2}{\nu^*\xi_t})) - \Phi(d_2(\frac{\kappa_1}{\nu^*\xi_t}))] \}}_{\text{risk seeking}} \\ &+ \underbrace{\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} c_3 \times \frac{\Phi'(d_1(\frac{\kappa_2}{\nu^*\xi_t}))}{\Phi'(d_2(\frac{\kappa_2}{\nu^*\xi_t}))} \times [1 - \Phi(d_2(\frac{\kappa_2}{\nu^*\xi_t}))]}_{\text{risk seeking}} \\ &\triangleq X_t^D + X_t^A + X_t^R. \end{aligned} \quad (29)$$

The optimal amount allocated in the risky asset, i.e., the optimal portfolio π^* is

$$\begin{aligned} \pi_t^* &= \underbrace{\frac{(\sigma^T)^{-1}\theta}{1-\gamma_1} X_t^*}_{\text{Merton term}} \\ &+ \underbrace{\frac{e^{-r(T-t)}}{\sqrt{T-t}} \frac{(\sigma^T)^{-1}\theta}{\|\theta\|_2} \{ [(1+c_1) \frac{B_1}{\alpha_1} - \underline{x}] \Phi'(d_1(\frac{\kappa_1}{\nu^*\xi_t})) + [(1+c_3) \frac{\beta(1-\omega)B_2+B_1}{\alpha_2} - (1+c_1) \frac{B_1}{\alpha_1}] \Phi'(d_1(\frac{\kappa_2}{\nu^*\xi_t})) \}}_{\text{risk seeking}} \\ &- e^{-r(T-t)} \frac{B_1}{\alpha_1} [\Phi(d_1(\frac{\kappa_2}{\nu^*\xi_t})) - \Phi(d_1(\frac{\kappa_1}{\nu^*\xi_t}))] \\ &+ \underbrace{\beta(1-\omega)B_2 + B_1 \frac{1}{\alpha_2 [1-\Phi(d_1(\frac{\kappa_2}{\nu^*\xi_t}))]} }_{\text{loss aversion}} \\ &- e^{-r(T-t)} \underbrace{\frac{(\sigma^T)^{-1}\theta}{1-\gamma_1} \{ \underline{x} \Phi(d_1(\frac{\kappa_1}{\nu^*\xi_t})) \}}_{\text{first-order risk aversion}} \\ &\triangleq \pi_t^{(1)} + \pi_t^{(2)} + \pi_t^{(3)} + \pi_t^{(4)}. \end{aligned} \quad (30)$$

(3) If $B_1 < \Theta(B_2)$, and $\kappa_1 \leq \kappa_2$, then the optimal terminal wealth X_T^* is

$$X_T^* = \underline{x} \mathbb{1}_{\{\nu^*\xi_T \geq \kappa_3\}} + (c_4 (\frac{\nu^*\xi_T}{\kappa_3})^{\frac{1}{\gamma_1-1}} + 1) \frac{B_1 + \beta(1-\omega)B_2}{\alpha_2} \mathbb{1}_{\{\nu^*\xi_T < \kappa_3\}}, \quad (31)$$

and the optimal wealth amount at time t is

$$\begin{aligned} X_t^* &= \xi_t^{-1} \mathbb{E}[\xi_T X_T^* | \mathcal{F}_t] \\ &= e^{-r(T-t)} \underbrace{\underline{x} \Phi(d_1(\frac{\kappa_3}{\nu^*\xi_t}))}_{\text{first-order risk aversion}} \\ &+ e^{-r(T-t)} \underbrace{\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} [1 - \Phi(d_1(\frac{\kappa_3}{\nu^*\xi_t}))]}_{\text{loss aversion}} \\ &+ e^{-r(T-t)} \underbrace{\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} c_4 \times \frac{\Phi'(d_1(\frac{\kappa_3}{\nu^*\xi_t}))}{\Phi'(d_2(\frac{\kappa_3}{\nu^*\xi_t}))} \times [1 - \Phi(d_2(\frac{\kappa_3}{\nu^*\xi_t}))]}_{\text{risk seeking}} \\ &\triangleq X_t^D + X_t^A + X_t^R. \end{aligned} \quad (32)$$

(29) The optimal amount allocated in the risky asset, i.e., the

optimal portfolio π^* is

$$\begin{aligned}
\pi_t^* &= \underbrace{\frac{(\sigma^T)^{-1}\theta}{1-\gamma_1} X_t^*}_{\text{Merton term}} \\
&+ \underbrace{\frac{e^{-r(T-t)}}{\sqrt{T-t}} \frac{(\sigma^T)^{-1}\theta}{\|\theta\|_2} \left[(1+c_4) \frac{\beta(1-\omega)B_2+B_1}{\alpha_2} - \underline{x} \right]}_{\text{loss aversion}} \\
&+ \underbrace{\frac{\Phi'(d_1(\frac{\kappa_3}{\nu^*\xi_t}))}{\alpha_2}}_{\text{risk seeking}} \\
&- \underbrace{\frac{\beta(1-\omega)B_2+B_1}{\alpha_2} [1-\Phi(d_1(\frac{\kappa_3}{\nu^*\xi_t}))]}_{\text{loss aversion}} \\
&- \underbrace{e^{-r(T-t)} \frac{(\sigma^T)^{-1}\theta}{1-\gamma_1} \underline{x} \Phi(d_1(\frac{\kappa_3}{\nu^*\xi_t}))}_{\text{first-order risk aversion}} \\
&\triangleq \pi_t^{(1)} + \pi_t^{(2)} + \pi_t^{(3)} + \pi_t^{(4)}.
\end{aligned} \tag{33}$$

where $d_1(z) := \frac{1}{-\|\theta\|\sqrt{T-t}}(\log(z) + (r - \frac{\|\theta\|^2}{2})(T-t))$, $z > 0$, $d_2(z) := d_1(z) - \frac{\|\theta\|\sqrt{T-t}}{1-\gamma_1}$, ν^* is a unique positive number that satisfies $\mathbb{E}[\xi_T X_T^*] = x_0$, and $\Phi(\cdot)$ is the standard normal cumulative distribution function.

As a direct implication of the unified formula of PHARA utility in (2), in those cases, X_t^* and π_t^* contains three terms. First, in X_t^* , X_t^D comes from the non-differentiable point $\underline{x}$ of the utility function, so it is called *first-order risk aversion term*, which leads to the $\pi_t^{(4)}$ in π_t^* . X_t^A comes from the loss-averse reference level of the non-linear part of the piecewise utility, so it is called *loss aversion term*. $\pi_t^{(3)}$ has the same name with the same reason. X_t^R is called the *risk seeking term* since it leads to $\pi_t^{(2)}$, which comes from the linear part of the piecewise utility. As in (10), the non-concave part of utility can lead to dramatic increases in volatility. Figure 2 illustrates the risk-seeking property on the interval of the linear part of the utility in example 1 (i.e. (a_0, a_1) and (a_2, a_3)).

IV. PROBLEM 2: AN S-SHAPED UTILITY AND A CONTRACT AND MARKET-BASED VAR CONSTRAINT

A. A non-concave utility maximization problem under a stochastic Value-at-Risk constraint

In this subsection, we formulate a constrained utility optimization problem with stochastic benchmarks:

$$\begin{aligned}
&\max_{\pi \in \mathcal{V}} \mathbb{E}[U(X_T)] \\
&\text{subject to } \mathbb{E}[\xi_T X_T] \leq x_0, \\
&\mathbb{P}(\Theta(X_T) \geq B_3) \geq 1 - \alpha.
\end{aligned} \tag{34}$$

where the objective function is the same as Problem 10, $\mathbb{P}(\Theta(X_T) \geq B_3) \geq 1 - \alpha$ is the VaR constraint and $\Theta(X_T)$ is the payoff of the manager.

In the previous literature, the Value-at-Risk constraint is used widely as a constraint in the optimal portfolio selection to manage risk. (11) are the first ones to directly embed risk

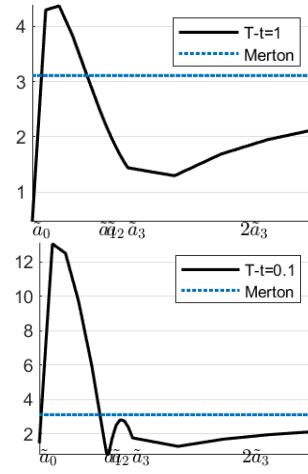


Fig. 2: The optimal portfolio with the setting in example 1 (y-axis: ‘Risky Investment Percentage’ means the optimal percentage invested in the risky asset $\frac{\pi_t^*}{X_t^*}$, x-axis: ‘Wealth’ means X_t^*)

management objectives into a utility maximizing problem. However, they only consider the problem under general state-independent preferences. That is, the benchmark B_3 is usually taken as a constant value L , which is the level of return under a given confidence interval. A natural thinking is that the benchmark B_3 can be higher (lower) when the market condition is good (bad), which is state-dependent and stochastic. (12) discusses the general thinking of state-dependence benchmarks and use the stochastic VaR benchmarks to solve the optimal terminal wealth of a constrained utility optimization problem. (3) investigate a VaR-constrained problem with S-shape utility and gives an algorithm to solve the Lagrange multiplier and optimal terminal wealth.

Following their works, we propose the optimal strategy for the utility optimization problem under the following setting: market-based VaR benchmarks, S-shape utility and incentive contract. We are interested in the case that a hedge fund manager determine her optimal strategy based on her payoff and her own risk preference. Instead of considering the worst return of the fund asset in a confident interval, she considers her own payoff with a VaR constraint. Moreover, she has a stochastic benchmark of the VaR constraint, which instantaneously depends on the market conditions.

To formulate the problem, we first consider the stochastic VaR benchmark B_3 in the simplest case (where L_i and w are constants):

$$B_3 = L_1 + (L_2 - L_1) \mathbb{1}_{\{W_T \geq w\}}$$

B_3 is higher when the market condition is better ($W_T \geq w$). The constant w is chosen as a threshold of a good market condition, below which the market condition is worse. Besides, we require $L_1 < L_2$ since the manager will have a higher expectation of the worst return B_3 when the market condition is good ($B_3(w) = L_2$). In consideration of numerical results, we require $\Theta(\underline{x}) < B_3 < B_1 < \Theta(B_2)$.

We use the lagrange multiplier and martingale method to solve the problem 34. Define a dual utility function

$$U^\mu(x, (b_1, b_2, b_3)) = \begin{cases} U(x, (b_1, b_2)) + \mu \mathbb{1}_{\{\Theta(x) \geq b_3\}}, & x \leq \underline{x}, \\ -\infty, & x < \underline{x}. \end{cases} \quad (35)$$

where $\mu \geq 0$ is a Lagrange multiplier to be determined. Denote $b = (b_1, b_2, b_3)$ and $B = (B_1, B_2, B_3)$. Consider the following revised problem without VaR constraint:

$$\begin{aligned} \max_{X_T \in \mathcal{A}} \mathbb{E}[U^\mu(X_T, B)] \\ \text{s.t. } \mathbb{E}[\xi_T X_T] \leq x_0. \end{aligned} \quad (36)$$

B. Solving the optimization problem 36 without VaR constraint

We first consider the case when $B_3 < B_1$, $B_1 < \Theta(B_2)$ and $\kappa_1 > \kappa_2$. Under the preceding setting, we have the following result.

Lemma 2. (1) *There exists a tangent point $(1 + c_5) \frac{B_1}{\alpha_1} \in (\frac{B_1}{\alpha_1}, B_2]$ deduced from $(\Theta^{-1}(B_3), U(\Theta^{-1}(B_3)))$ to $U|_{(\frac{B_1}{\alpha_1}, B_2]}$, which satisfies*

$$\frac{U^\mu((1 + c_5) \frac{B_1}{\alpha_1}) - U^\mu(\Theta^{-1}(B_3))}{(1 + c_5) \frac{B_1}{\alpha_1} - \underline{x}} = U^{\mu'}((1 + c_5) \frac{B_1}{\alpha_1}). \quad (37)$$

Define $y_1(B)$ as the slope of the tangent line:

$$y_1(B) \triangleq U^\mu((1 + c_5) \frac{B_1}{\alpha_1}). \quad (38)$$

(2) Define $y_2^\mu(B)$ as the slope of the line joining $(\underline{x}, U^\mu(\underline{x}))$ and $(\Theta^{-1}(B_3), U^\mu(\Theta^{-1}(B_3)) + \mu)$:

$$y_2^\mu(B) \triangleq \frac{U^\mu(\Theta^{-1}(B_3)) - U^\mu(\underline{x})}{\Theta^{-1}(B_3) - \underline{x}}. \quad (39)$$

(3) *There exists a tangent point $(1 + c_6) \frac{B_1}{\alpha_1} \in (\frac{B_1}{\alpha_1}, B_2]$ deduced from $(\underline{x}, U^\mu(\underline{x}))$ to $U^\mu|_{(\frac{B_1}{\alpha_1}, B_2]}$, which satisfies*

$$\frac{U^\mu((1 + c_6) \frac{B_1}{\alpha_1}) - U^\mu(\underline{x})}{(1 + c_6) \frac{B_1}{\alpha_1} - \underline{x}} = U^{\mu'}((1 + c_6) \frac{B_1}{\alpha_1}). \quad (40)$$

Define $y_3^\mu(B)$ as the slope of the tangent line:

$$y_3^\mu(B) \triangleq U^{\mu'}((1 + c_6) \frac{B_1}{\alpha_1}). \quad (41)$$

where $U^\mu(x) := U^\mu(x, B)$.

Theorem 2. For every $x_0 \geq \underline{x}$, Problem 36 admits a unique finite optimal solution $X_B^\mu(\nu(\mu)\xi)$ with ν as a function of μ . There are two cases:

(1) if $y_1(b) \leq y_2^\mu(b)$, the optimal solution to the problem 36 is

$$X_b^{\textcircled{1}\mu}(y) = \begin{cases} [c_3(\frac{y}{\kappa_2})^{\frac{1}{\gamma_1-1}} + 1] \frac{\beta(1-\omega)B_2 + B_1}{\alpha_2}, & y < \kappa_2, \\ [c_5(\frac{y}{y_1(b)})^{\frac{1}{\gamma_1-1}} + 1] \frac{B_1}{\alpha_1}, & \kappa_2 \leq y < y_1(b), \\ \Theta^{-1}(B_3), & y_1(b) \leq y < y_2^\mu(b), \\ \underline{x}, & y_2^\mu(b) \leq y \end{cases} \quad (42)$$

(2) if $y_1(b) \geq y_2^\mu(b)$, the optimal solution to the problem 36 is

$$X_b^{\textcircled{2}\mu}(y) = \begin{cases} [c_3(\frac{y}{\kappa_2})^{\frac{1}{\gamma_1-1}} + 1] \frac{\beta(1-\omega)B_2 + B_1}{\alpha_2}, & y < \kappa_2, \\ [c_1(\frac{y}{y_3^\mu(b)})^{\frac{1}{\gamma_1-1}} + 1] \frac{B_1}{\alpha_1}, & \kappa_2 \leq y < y_3^\mu(b), \\ \underline{x}, & y_3^\mu(b) \leq y \end{cases} \quad (43)$$

The Lagrange multiplier μ^* comes from $\mathbb{P}(\Theta(X_T) \geq B_3) \geq 1 - \alpha$, but it may not exist. If μ^* exist, $X_T^* = X_B^\mu(\nu(\mu^*)\xi_T)$ gives a finite optimal solution of Problem 34, with ν satisfying $\mathbb{E}[\xi_T X_T] = x_0$.

Next, we derive the optimal wealth process X_t^* based on Theorem 2. Because the random variable W_T determine the value of B_3 and μ^* , hence the relationship of $y_1(B)$ and $y_2^\mu(B)$ can be determined based on given values of parameters. We split X_T^* into four terms:

$$\begin{aligned} X_T^* &= \mathbb{1}_{\{W_T > w\}} X_B^{\textcircled{1}\mu}(\nu^* \xi_T) \mathbb{1}_{\{y_1(B) < y_2^\mu(B)\}} \\ &\quad + \mathbb{1}_{\{W_T \leq w\}} X_B^{\textcircled{1}\mu}(\nu^* \xi_T) \mathbb{1}_{\{y_1(B) < y_2^\mu(B)\}} \\ &\quad + \mathbb{1}_{\{W_T > w\}} X_B^{\textcircled{2}\mu}(\nu^* \xi_T) \mathbb{1}_{\{y_1(B) \geq y_2^\mu(B)\}} \\ &\quad + \mathbb{1}_{\{W_T \leq w\}} X_B^{\textcircled{2}\mu}(\nu^* \xi_T) \mathbb{1}_{\{y_1(B) \geq y_2^\mu(B)\}} \\ &\triangleq X_T^1 + X_T^2 + X_T^3 + X_T^4. \end{aligned} \quad (44)$$

Theorem 3. There are four cases of the optimal wealth process X_t^* :

(1) if $y_1((B_1, B_2, L_1)) < y_2^\mu((B_1, B_2, L_1))$ and $y_1((B_1, B_2, L_2)) < y_2^\mu((B_1, B_2, L_2))$, then the optimal wealth process is

$$X_t^* = \xi_t^{-1} \mathbb{E}[\xi_T (X_T^1 + X_T^2) | \mathcal{F}_t]. \quad (45)$$

(2) if $y_1((B_1, B_2, L_1)) \geq y_2^\mu((B_1, B_2, L_1))$ and $y_1((B_1, B_2, L_2)) < y_2^\mu((B_1, B_2, L_2))$, then the optimal wealth process is

$$X_t^* = \xi_t^{-1} \mathbb{E}[\xi_T (X_T^1 + X_T^4) | \mathcal{F}_t]. \quad (46)$$

(3) if $y_1((B_1, B_2, L_1)) < y_2^\mu((B_1, B_2, L_1))$ and $y_1((B_1, B_2, L_2)) \geq y_2^\mu((B_1, B_2, L_2))$, then the optimal wealth process is

$$X_t^* = \xi_t^{-1} \mathbb{E}[\xi_T (X_T^2 + X_T^3) | \mathcal{F}_t]. \quad (47)$$

(4) if $y_1((B_1, B_2, L_1)) \geq y_2^\mu((B_1, B_2, L_1))$ and $y_1((B_1, B_2, L_2)) \geq y_2^\mu((B_1, B_2, L_2))$, then the optimal wealth process is

$$X_t^* = \xi_t^{-1} \mathbb{E}[\xi_T (X_T^3 + X_T^4) | \mathcal{F}_t]. \quad (48)$$

where with X_T^i not equal to zero for some W_T , the solution is given by:

$$\begin{aligned}
X_t^1 &= \xi_t^{-1} \mathbb{E}[\xi_T(X_T^1) | \mathcal{F}_t] \\
&= e^{-r(T-t)} \{ \underline{x} [\Phi(d_1(\frac{y_2^\mu(B)}{\nu^* \xi_t})) - \Phi(\frac{w - W_t}{\sqrt{T-t}})] + \\
&\quad + \Theta^{-1}(B_3) [\Phi(d_1(\frac{y_1(B)}{\nu^* \xi_t})) \\
&\quad - \Phi(\max\{d_1(\frac{y_2^\mu(B)}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}}\})] + \\
&\quad + \frac{\beta(1-\omega)B_2 + B_1}{\alpha_2} [1 - \Phi(\max\{d_1(\frac{\kappa_2}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}}\})] \\
&\quad + \frac{B_1}{\alpha_1} [\Phi(d_1(\frac{\kappa_2}{\nu^* \xi_t})) - \Phi(\max\{d_1(\frac{y_1(B)}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}}\})] + \\
&\quad + c_3 \frac{\beta(1-\omega)B_2 + B_1}{\alpha_2} \times \frac{\Phi'(d_1(\frac{\kappa_2}{\nu^* \xi_t}))}{\Phi'(d_2(\frac{\kappa_2}{\nu^* \xi_t}))} \\
&\quad \times [1 - \Phi(\max\{d_2(\frac{\kappa_2}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}} + \frac{\gamma_1}{\gamma_1 - 1} \|\theta\| \sqrt{T-t}\})] \\
&\quad + c_5 \frac{B_1}{\alpha_1} \times \frac{\Phi'(d_1(\frac{y_1(B)}{\nu^* \xi_t}))}{\Phi'(d_2(\frac{y_1(B)}{\nu^* \xi_t}))} \\
&\quad \times [\Phi(d_2(\frac{\kappa_2}{\nu^* \xi_t})) - \\
&\quad \Phi(\max\{d_2(\frac{y_1(B)}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}} + \frac{\gamma_1}{\gamma_1 - 1} \|\theta\| \sqrt{T-t}\})] + \} \\
&\quad (49)
\end{aligned}$$

$$\begin{aligned}
X_t^2 &= \xi_t^{-1} \mathbb{E}[\xi_T(X_T^2) | \mathcal{F}_t] \\
&= e^{-r(T-t)} \{ \underline{x} [\min\{\Phi(d_1(\frac{y_2^\mu(B)}{\nu^* \xi_t})), \Phi(\frac{w - W_t}{\sqrt{T-t}})\}] \\
&\quad + \Theta^{-1}(B_3) [\Phi(\min\{d_1(\frac{y_1(B)}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}}\}) - \\
&\quad \Phi(d_1(\frac{y_2^\mu(B)}{\nu^* \xi_t}))] + \\
&\quad + \frac{\beta(1-\omega)B_2 + B_1}{\alpha_2} [\Phi(\frac{w - W_t}{\sqrt{T-t}}) - \Phi(d_1(\frac{\kappa_2}{\nu^* \xi_t}))] + \\
&\quad + \frac{B_1}{\alpha_1} [\Phi(\min\{d_1(\frac{\kappa_2}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}}\}) - \Phi(d_1(\frac{y_1(B)}{\nu^* \xi_t}))] + \\
&\quad + c_3 \frac{\beta(1-\omega)B_2 + B_1}{\alpha_2} \times \frac{\Phi'(d_1(\frac{\kappa_2}{\nu^* \xi_t}))}{\Phi'(d_2(\frac{\kappa_2}{\nu^* \xi_t}))} \\
&\quad \times [\Phi(\frac{w - W_t}{\sqrt{T-t}} + \frac{\gamma_1}{\gamma_1 - 1} \|\theta\| \sqrt{T-t}) - \Phi(d_2(\frac{\kappa_2}{\nu^* \xi_t}))] + \\
&\quad + c_5 \frac{B_1}{\alpha_1} \times \frac{\Phi'(d_1(\frac{y_1(B)}{\nu^* \xi_t}))}{\Phi'(d_2(\frac{y_1(B)}{\nu^* \xi_t}))} \\
&\quad \times [\Phi(\min\{d_2(\frac{\kappa_2}{\nu^* \xi_t}), \frac{w - W_t}{\sqrt{T-t}} + \frac{\gamma_1}{\gamma_1 - 1} \|\theta\| \sqrt{T-t}\}) \\
&\quad - \Phi(d_2(\frac{y_1(B)}{\nu^* \xi_t}))] + \} \\
&\quad (50)
\end{aligned}$$

The terms X_t^3 and X_t^4 can be derive similarly.

Note that the optimal wealth process X_t^* depends on the

benchmark w in B_3 , and the state of market W_t at time t . If we let $w \rightarrow -\infty$, B_3 is equal to L_2 almost surely since W_t is bounded on $[0, T]$. In this case, the solution X_t^* goes back to the form from the unified formula of PHARA utility (cf. (2)).

Building upon the unified formula for piecewise hyperbolic absolute risk aversion (PHARA) utilities as outlined by (2), the portfolio optimization for hedge fund managers under an incentive contract is further nuanced through the incorporation of a stochastic VaR constraint relative to a market benchmark.

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